

A simple method for studying low-gravity sloshing frequencies

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A general method is presented to compute low-gravity sloshing frequencies of liquids in spherical and cylindrical tanks. The key idea is to use two independent coordinate systems for the interface displacement η and liquid velocity potential Φ , respectively, which permit the contact-angle condition and the normal-velocity condition on the tank wall to be satisfied exactly. The functions η and Φ are then expanded in terms of independent complete sets of spatial functions. When these are substituted into the liquid free-surface conditions and the latter are projected onto the basis used for η , we obtain a standard matrix eigenvalue problem $Az = \lambda Bz$, where the eigenvalue λG is either the natural frequency of oscillation or its square. It is shown that the method easily applies for all the cases of interest. In particular, it is shown that, for the flat-meniscus case at infinite Bond number, the present results generally agree with the accurate results of McIver. A useful by-product of the method is that estimates of the lowest frequencies, requiring quadratures alone, can be obtained in simple closed form. Finally, it is shown that the method can be extended to the arbitrary Bond number case, the only extra complication being the need to determine the distorted static meniscus.

Keywords: microgravity; sloshing; natural frequencies; contact angle; capillarity

1. Introduction

The theory of small-amplitude oscillations of a liquid mass with a flat free surface under gravity is classical and is discussed, for example, in ch. VIII and IX of Lamb (1945). When the container geometry is simple and the flat free surface is one of the coordinate surfaces, the linear problem can be easily solved for the required natural frequencies of oscillation of the liquid mass. However, if the container geometry is not simple or if the free surface is not flat, as in the low-gravity case, even the linear problem is no longer trivial and the estimation of the natural frequencies becomes a complicated task. It is for this reason that over the years there has been a stream of literature on the subject, of which a representative sample is Budiansky (1960), Chu (1964, 1970), Kuttler & Sigillito (1984), Myshkis *et al.* (1987), McIver (1989) and Utsumi (2000). In view of the very great practical importance of the subject it is likely that work will continue in this classical area for some time to come.

The applications are many. Terrestrial applications include sloshing in the tanks of fluid-bearing road vehicles and in the tanks of ships. Here gravity dominates, the liquid free surface is almost flat and the main complexity is the geometry of the tank.

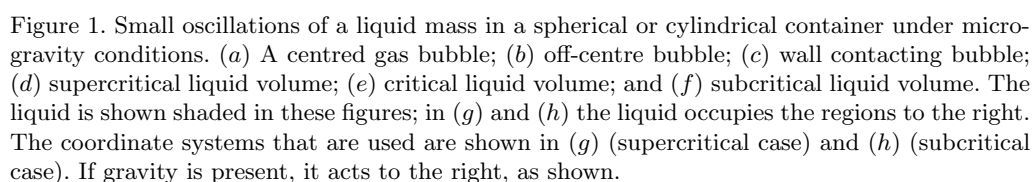
On the other hand, in space applications, which have been the prime motivation for the present work, the container geometry is often quite simple. Now, however, in a low-gravity environment the liquid free surface is highly curved and surface-tension effects dominate. Whereas in the flat-interface case it is possible to devise analytical methods of great accuracy (McIver 1989), at least for simple container geometries, this does not seem to be the case in the low-gravity, i.e. low-Bond-number (low- B), situation.

There are two main difficulties in the calculation of the natural frequencies of oscillation: firstly, the representation or computation of the potential and secondly, the satisfaction of the boundary conditions on the interface. In most of the early work the advantages of the complex potential in two dimensions or the existence of appropriate Green functions were exploited to yield integral equations for the unknowns (Budiansky 1960; Chu 1964; Kuttler & Sigillito 1984). Another line of attack has been the reduction of the problem to a variational problem (Myshkis *et al.* 1987; Utsumi 2000). For simple container geometries and a flat interface, use can be made of special coordinate systems in which the interface is a coordinate surface. Using such an approach, McIver (1989) computed the natural frequencies of liquids in cylindrical and spherical containers filled to arbitrary depths when surface tension was absent (i.e. when $B = \infty$). It appears that McIver's results are the most accurate available for these geometries under these conditions. The method used, however, does not seem to carry over to the arbitrary-Bond-number case.

In most of the above approaches, the natural frequencies follow from the solution of a matrix eigenvalue problem of the type $Az = \lambda Bz$, where A and B are matrices which depend on the parameters of the problem and λ is an eigenvalue. However, in none of the approaches that we have seen are there explicit formulae for the matrix elements, since the formulations are so complicated. The present work was motivated by the need for a simple and reasonably accurate method to compute low-gravity modal frequencies of liquid masses in containers of simple geometry with highly curved liquid free surfaces.

We are particularly concerned with partly filled, spherical satellite fuel tanks. One of the difficulties in a low-gravity environment is that the location of the liquid fuel in the tank may take a long time to stabilize after a manoeuvre. In fact, at zero g , it is possible in principle for the liquid to make no contact with the walls of the tank. Once contact is made and the liquid attaches to the tank wall, it must have a spherical free surface and satisfy the contact-angle condition at the wall. In figure 1 we show, for zero- g conditions, six cases of interest, (a)–(f), that need to be considered. Note that, in the figure what is shown in each case is a cross-section through the centre of the spherical tank. In figure 1a the liquid is attached to the wall with a central gas bubble; the bubble is off-centre in figure 1b, while in figure 1c the bubble just touches the wall (the contact angle α is zero). For a given α , there is a critical volume V_c of the liquid mass for which the interface, defined to be the surface separating the liquid from the gas, is flat at zero g (figure 1e). When the volume of fuel is greater than V_c , the volume is supercritical and the liquid–gas interface is concave, as shown in figure 1d. Finally, when the volume is less than V_c , we say it is subcritical and the interface is convex as shown in figure 1f.

Our purpose is to devise a simple method that will work for all of the above cases and that can be extended to other simple geometries. Although the method is motivated by low-gravity space applications, we will show that it can be extended



We shall assume throughout that the unperturbed liquid mass does not occupy the centre of the tank—a limitation that is not serious for the applications that we have in mind. However, we shall later indicate how this restriction can be removed.

2. The natural frequencies of a supercritical volume of liquid in a spherical container

In order to fix ideas, let us consider the supercritical case sketched in figure 1*d* and, in greater detail, in figure 1*g*. Let ρ and σ be the density and surface tension of the inviscid fluid, R be the radius of the tank and g the acceleration due to gravity. Let all lengths be non-dimensionalized by R , the time by $T = \sqrt{\rho R^3/\sigma}$, velocities by R/T and the pressure by σ/R . In a suitable coordinate frame $\mathbf{x}$, let $\zeta(\mathbf{x};t)$ represent the liquid free surface and let $\Phi(\mathbf{x};t)$ represent the velocity potential. The oscillations of the liquid are then governed by two equations that ensure the existence of the interface and the satisfaction of the Laplace–Young condition across it,

$$\zeta_{,t} + (\mathbf{u} \cdot \nabla)\zeta = u_n, \quad (2.1 a)$$

$$\Phi_{,t} + \frac{1}{2}|\mathbf{u}|^2 - B\zeta \cos \theta = \kappa + \gamma, \quad (2.1 b)$$

which have to be satisfied at each instant on the liquid free surface. Here $\mathbf{u} = \nabla\Phi$ is the liquid velocity, u_n is its component normal to the interface, θ is the angle between the direction of gravity and the normal to the interface, κ is the curvature of the interface, $B = \rho g R^2/\sigma$ is the Bond number and γ is a constant determined by the interface datum. Subscripts following a comma indicate, here and in the following, differentiation with respect to them. The general formula for the curvature in spherical polars is given in Appendix B.

On the tank wall, assumed to be stationary in this frame, we require the normal component of the velocity to vanish. We then need a condition on the interface at the contact line. We shall assume that the interface makes an angle with the tank wall α , the constant contact angle, throughout the motion. It is well known that this model for the liquid–solid–gas contact line is not a very good one for some triplets (see, for example, Dussan 1979). Cases exist where the contact line can exhibit hysteresis, where viscous forces play a dominant role near the line, etc. The issue is a complex one which does not permit a simple resolution. In the specific case of drop spreading, Hocking (1992), in comparing the use of a dynamic contact angle depending on the contact-line velocity and one equal to the static value, came to the conclusion that the latter was to be preferred. The position taken here is that the dynamic contact angle may be taken to be equal to its static value α , at least as a good approximation in many cases of interest.

The key idea in the present method is to use *two* coordinate systems, one for the interface and one for the potential. This will permit, as we shall show, the possibility of representing both η and Φ analytically and satisfying the contact angle and no-penetration conditions exactly. That this is in general difficult to achieve can be seen from the fact that, in as recent a work as Utsumi (2000), both the contact-angle condition and the no-penetration condition are only satisfied approximately. In brief, the method consists of the following steps:

- (i) use two convenient coordinate systems $\mathbf{x}$ and $\hat{\mathbf{x}}$ for ζ and Φ , respectively;
- (ii) choose sets of basis functions, complete on the appropriate intervals, which permit satisfaction of the tank-wall conditions and expand ζ and Φ in terms of them;

- (iii) project the linearized interface conditions onto the basis used for ζ to obtain a matrix eigenvalue problem of the type $Az = \lambda Bz$, where λ is either the frequency or its square;
- (iv) solve the latter for the frequencies.

While this methodology applies for all the cases of interest, a few of the details will be shown here for the supercritical case.

Consider figure 1*g* and, to keep matters simple initially, assume no gravity. The quiescent liquid now occupies a domain bounded by two spherical surfaces, the wall of the container and the liquid free surface. Let the radius of curvature of the free surface be r_0 , which is a function of the liquid fill volume fraction V and α alone. In fact

$$V = 0.75\left\{\frac{2}{3} - \cos \hat{\theta}_0 + \frac{1}{3} \cos^3 \hat{\theta}_0 - r_0^3\left(\frac{2}{3} - \cos \theta_0 + \frac{1}{3} \cos^3 \theta_0\right)\right\},$$

where $r_0 = \sin \hat{\theta}_0 / \sin(\hat{\theta}_0 - \alpha)$ and $\theta_0 = \hat{\theta}_0 - \alpha$.

Choose spherical polar coordinate systems (r, θ, ϕ) and $(\hat{r}, \hat{\theta}, \hat{\phi})$, as shown for ζ and Φ , respectively, where θ and $\hat{\theta}$ are the polar angles. Note that, since the liquid only occupies the crescent-shaped region in figure 1*g*, it is sufficient for the range of $\hat{\theta}$ to be $[0, \hat{\theta}_0]$. If (2.1 *a*) and (2.1 *b*) are now linearized for small oscillations $\zeta = r_0 + \eta(\theta, \phi; t)$ about the static interface, using the unhatted coordinate system we have

$$\eta_{,t} = u_r, \quad (2.2 \ a)$$

$$\Phi_{,t} = -(1/r_0^2)\{2\eta + \cot \theta \eta_{,\theta} + \eta_{,\theta\theta} + \csc^2 \theta \eta_{,\phi\phi}\}, \quad (2.2 \ b)$$

as the governing equations. The expression on the right-hand side of (2.2 *b*) is the first-order correction to the interface curvature and is derived in Appendix B as equation (B 4). Note that to this order the constant γ in (2.1 *b*) is just minus the curvature to leading order $2/r_0$ and hence drops out of (2.2 *b*).

In order to find the natural frequencies of oscillation, we now assume the dependent variables to be harmonic in time and expand, for a symmetric azimuthal wavenumber mode m , their spatial components in terms of spherical harmonics (Lebedev 1965):

$$\eta(\theta, \phi; t) = \cos \omega t \cos m\phi \sum_{j=1}^{\infty} a_{mj} P_{\alpha_j}^m(\cos \theta), \quad (2.3 \ a)$$

$$\Phi(\hat{r}, \hat{\theta}, \hat{\phi}; t) = \sin \omega t \cos m\hat{\phi} \sum_{j=1}^{\infty} b_{mj} \left(\hat{r}^{\beta_j} + \frac{\beta_j}{\beta_j + 1} \hat{r}^{-\beta_j - 1} \right) P_{\beta_j}^m(\cos \hat{\theta}). \quad (2.3 \ b)$$

Here ω is the natural frequency, which is to be determined, as are the scalars a_{mj} and b_{mj} . Note from (2.3 *b*) that Φ is harmonic and that the normal velocity on the tank wall, $\hat{r} = 1$, vanishes. We now choose the scalars $\{\alpha_j, j = 1, 2, \dots\}$ and $\{\beta_j, j = 1, 2, \dots\}$ to be those that satisfy

$$\left. \frac{d}{d\theta} \{P_{\alpha_j}^m(\cos \theta)\} \right|_{\theta=\theta_0} = 0, \quad (2.4 \ a)$$

$$P_{\beta_j}^m(\cos \hat{\theta})|_{\hat{\theta}=\hat{\theta}_0} = 0. \quad (2.4 \ b)$$

Note that (2.4 *a*) ensures that the contact-angle condition is satisfied exactly. To see this, we first observe that the contact angle at any point on the contact line must

equal the angle between the normal to the interface at that point and the normal to the spherical container. The spherical shell is given by $\hat{f}(\hat{r}, \hat{\theta}, \hat{\phi}) = \hat{r} - 1 = 0$, while the interface is given by

$$\begin{aligned} f(r, \theta, \phi) &= r - \zeta \\ &= r - r_0 - \eta(\theta, \phi; t) \\ &= 0. \end{aligned}$$

If we now find the normals and take their scalar products, keeping in mind that $\mathbf{e}_{\hat{r}} \cdot \mathbf{e}_r = \cos \alpha$, while $\mathbf{e}_{\hat{r}} \cdot \mathbf{e}_{\phi} = 0$, we find that $\eta_{,\theta} = 0$ ensures that the contact-angle condition is satisfied. The condition (2.4 *b*) is dictated by the fact that we need to generate a complete set of functions on the appropriate interval consistent with the interface conditions.

We are now guaranteed that η and Φ are represented in terms of infinite sets of functions that are complete on the intervals $[0, \theta_0]$ and $[0, \hat{\theta}_0]$, respectively, and hence that our expansions (2.3) are valid. Noting that

$$u_r = u_{\hat{r}} \cos(\hat{\theta} - \theta) - u_{\hat{\theta}} \sin(\hat{\theta} - \theta), \quad (2.5 a)$$

$$u_{\theta} = u_{\hat{r}} \sin(\hat{\theta} - \theta) + u_{\hat{\theta}} \cos(\hat{\theta} - \theta), \quad (2.5 b)$$

equations (2.2 *a*) and (2.2 *b*), which hold on $r = r_0$, can now be written

$$\begin{aligned} -\omega \sum_{j=1}^{\infty} a_{mj} P_{\alpha_j}^m &= \sum_{j=1}^{\infty} b_{mj} \left\{ \hat{r}^{\beta_j-1} [\beta_j \cos(\hat{\theta} - \theta) P_{\beta_j}^m - \sin(\hat{\theta} - \theta) P_{\beta_j}^{m'}] \right. \\ &\quad \left. - \hat{r}^{-(\beta_j+2)} \left[\beta_j \cos(\hat{\theta} - \theta) P_{\beta_j}^m + \frac{\beta_j}{\beta_j + 1} \sin(\hat{\theta} - \theta) P_{\beta_j}^{m'} \right] \right\}, \end{aligned} \quad (2.6 a)$$

$$\omega \sum_{j=1}^{\infty} b_{mj} \left(\hat{r}^{\beta_j} + \frac{\beta_j}{\beta_j + 1} \hat{r}^{-\beta_j-1} \right) P_{\beta_j}^m = -\frac{1}{r_0^2} \sum_{j=1}^{\infty} a_{mj} (2 - \alpha_j(\alpha_j + 1)) P_{\alpha_j}^m, \quad (2.6 b)$$

where the arguments of the spherical harmonics $P_{\alpha_j}^m$ and $P_{\beta_j}^m$ are understood to be $\cos \theta$ and $\cos \hat{\theta}$, respectively, and the primes indicate differentiation with respect to $\hat{\theta}$. Note that the right-hand side of (2.6 *b*) follows from (2.2 *b*), (2.3 *a*) and the associated Legendre equation. We now obtain an infinite system of equations for the unknown scalars by projecting (2.6) onto the orthogonal basis $\{P_{\alpha_j}^m, j = 1, 2, \dots\}$ used to represent η . Multiplying the equations by $\sin \theta P_{\alpha_i}^m$ and integrating over $[0, \theta_0]$ we get, for $i = 1, 2, \dots$, the infinite system of algebraic equations

$$-\omega \gamma_i a_{mi} = \sum_{j=1}^{\infty} \mu_{ij} b_{mj}, \quad (2.7 a)$$

$$\omega \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = \frac{(\alpha_i + 2)(\alpha_i - 1)}{r_0^2} \gamma_i a_{mi}, \quad (2.7 b)$$

where

$$\gamma_i = \int_0^{\theta_0} \sin \theta (P_{\alpha_i}^m)^2 d\theta, \quad (2.8a)$$

$$\begin{aligned} \mu_{ij} = \int_0^{\theta_0} \left[\hat{r}^{\beta_j-1} \{ \beta_j \cos(\hat{\theta} - \theta) P_{\beta_j}^m - \sin(\hat{\theta} - \theta) P_{\beta_j}^{m'} \} \right. \\ \left. - \hat{r}^{-(\beta_j+2)} \left\{ \beta_j \cos(\hat{\theta} - \theta) P_{\beta_j}^m + \frac{\beta_j}{\beta_j + 1} \sin(\hat{\theta} - \theta) P_{\beta_j}^{m'} \right\} \right] \sin \theta P_{\alpha_i}^m d\theta, \end{aligned} \quad (2.8b)$$

$$\nu_{ij} = \int_0^{\theta_0} \left(\hat{r}^{\beta_j} + \frac{\beta_j}{\beta_j + 1} \hat{r}^{-\beta_j-1} \right) \sin \theta P_{\beta_j}^m P_{\alpha_i}^m d\theta. \quad (2.8c)$$

It is to be noted that, in the above integrals, $\hat{r}$ and $\hat{\theta}$ are known elementary functions of θ . In fact,

$$\hat{\theta} = \arctan \{ \sin \theta (\cos \theta - (l/r_0))^{-1} \}$$

and

$$\hat{r}(\hat{\theta}) = -l \cos \hat{\theta} + \sqrt{(l \cos \hat{\theta})^2 + (r_0^2 - l^2)}$$

on the interface. The coefficients a_{mi} can now be eliminated from equations (2.7) to get, for $i = 1, 2, \dots$, the system

$$\omega^2 \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = - \frac{(\alpha_i + 2)(\alpha_i - 1)}{r_0^2} \sum_{j=1}^{\infty} \mu_{ij} b_{mj}. \quad (2.9)$$

Thus, the problem has been reduced to a standard matrix eigenvalue problem $Az = \lambda Bz$, where $z^T = (b_{m1}, b_{m2}, \dots)$ and the eigenvalue λ is the square of the required natural frequency. In general, we obtain an infinite number of real positive eigenvalues for each m , which can be ordered in terms of increasing frequency, $\{\omega_{mi}, i = 1, 2, \dots\}$. From a practical point of view the lowest frequencies are of primary interest.

The matrix eigenvalue problem (2.9) for ω^2 can be solved numerically by standard routines (Press *et al.* 1996). All the sums are truncated to n terms, yielding in this way n eigenvalues for every $n \times n$ system solved. Although it is generally true that, as n is increased, the lower eigenvalues converge to their ultimate values, in most cases good accuracy can be obtained with very few terms; in fact, often the lowest two frequencies can be calculated to reasonable accuracy using just one or two spatial modes, i.e. with n equal to just one or two, the calculations then reducing to the use of closed-form formulae. This will be demonstrated in the examples that follow.

Figure 2 shows the results of typical calculations of the dependence on $\hat{\theta}_0$ of the lowest natural frequencies for crescent-shaped liquid masses of the type shown in figure 1*g*. Here $\hat{\theta}_0$ ranges from 145° to 45° and so the figures show how the natural frequencies change as the fuel depletes. The solid lines deal with the $m = 0$ modes, while the dashed lines deal with the first non-axisymmetric mode $m = 1$. In this

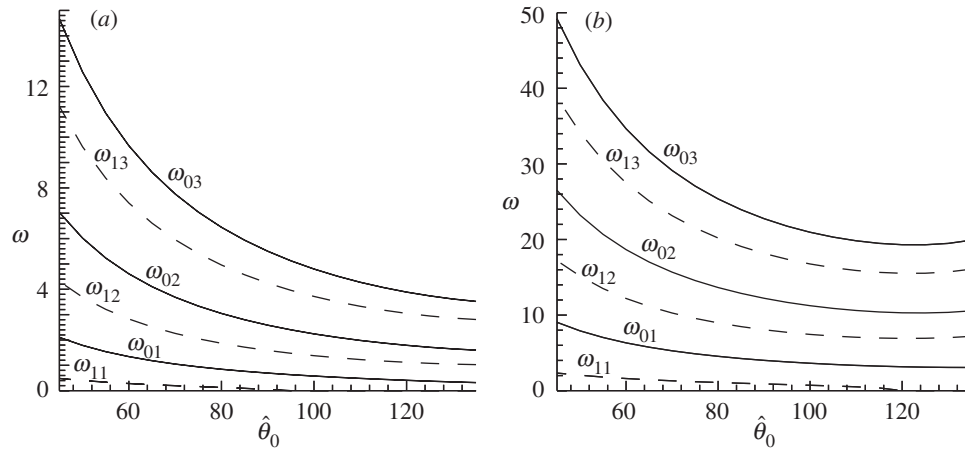


Figure 2. The variation of the lowest three natural frequencies with $\hat{\theta}_0$ for liquid in a spherical container. Bond number $B = 0$. (a) Contact angle $\alpha = 2^\circ$; (b) contact angle $\alpha = 30^\circ$.

Table 1. The dependence of the lowest natural frequencies on the number of modes n (The contact angle $\alpha = 30^\circ$. (a) $\hat{\theta}_0 = 45^\circ$; (b) $\hat{\theta}_0 = 135^\circ$)

	n	$m = 0$			$m = 1$		
		ω_{01}	ω_{02}	ω_{03}	ω_{11}	ω_{12}	ω_{13}
(a)	1	9.184	—	—	2.705	—	—
	2	9.062	27.274	—	2.515	18.806	—
	5	9.043	26.525	49.371	2.361	17.437	39.311
	10	9.040	26.481	49.300	2.362	17.461	39.342
(b)	1	3.287	—	—	0	—	—
	2	3.140	11.301	—	0	7.78	—
	5	3.065	10.529	19.408	0	7.286	16.199
	10	3.065	10.529	19.408	0	6.621	16.940

range for $\alpha = 2^\circ$ the frequencies uniformly increase with depletion; however, for $\alpha = 30^\circ$ this is true only for the fundamental. Also note that:

- (i) the frequencies for $m = 1$ are lower than the corresponding ones for $m = 0$ and this is usually the case;
- (ii) the fundamental seems to disappear for large $\hat{\theta}_0$ for some of the cases;
- (iii) the frequencies are higher for the higher contact angle case.

Since these facts result from a complicated interplay between the effects of liquid mass and liquid surface energy, there does not appear to be a simple explanation for these facts. It is of interest to see how the lowest frequencies converge typically to their ultimate values. For the $\alpha = 30^\circ$ case, table 1 shows how the lowest eigenvalues converge, at the extreme values of $\hat{\theta}_0$, as n is increased from 1 to 10. It will be

noted that the convergence for $m = 0$ is very rapid, while it is much slower for the asymmetric $m = 1$ mode; this is normally so.

From the above it can be seen that often we get good estimates for the lowest frequencies using very few terms in the series expansions. If only one spatial mode is used, the estimate for the lowest frequency is immediately

$$\omega_{m1} = \sqrt{K_1 \frac{\mu_{11}}{\nu_{11}}}, \quad (2.10)$$

where $K_i = -(\alpha_i + 2)(\alpha_i - 1)/r_0^2$. If we take two spatial modes, i.e. $n = 2$, we need only solve a quadratic to get estimates for the first two lowest frequencies

$$\omega_{m2}, \omega_{m1} = \frac{b \pm \sqrt{b^2 - 4ac}}{2a}, \quad (2.11)$$

where

$$a = \nu_{11}\nu_{22} - \nu_{21}\nu_{12}, \quad (2.12a)$$

$$b = K_1(\nu_{22}\mu_{11} - \nu_{21}\mu_{12}) + K_2(\nu_{11}\mu_{22} - \nu_{12}\mu_{21}), \quad (2.12b)$$

$$c = K_1K_2(\mu_{11}\mu_{22} - \mu_{12}\mu_{21}). \quad (2.12c)$$

Thus we have closed-form estimates for the lowest frequencies that only require quadratures. This is a useful bonus of the method.

We shall postpone an assessment of the accuracy of the method until § 3, when we will be able to compare our results with the very accurate results of McIver (1989), which are available in tabular form; thus errors resulting from the scanning of graphs of poor resolution can be avoided.

This section will now be concluded by pointing out how the effects of small Bond number can be incorporated into the analysis. Let us assume $0 < B \ll 1$; then the static meniscus will have a representation

$$\zeta(\theta, \phi) = r_0 + B\eta_s(\theta) + \dots, \quad (2.13)$$

where the second term, the correction to the purely spherical surface obtained in zero g , has to satisfy the equation

$$\eta_{s,\theta\theta} + \cot \theta \eta_{s,\theta} + 2\eta_s = r_0^3 B \cos \theta. \quad (2.14)$$

This can be easily integrated in closed form as shown in Appendix B. In order to find the natural frequencies we now write

$$\zeta(\theta, \phi; t) = r_0 + B\eta_s(\theta) + \eta(\theta, \phi; t) + \dots, \quad \eta \ll r_0 + B\eta_s,$$

and find that to leading order the interface conditions now read

$$\eta_{,t} + \frac{B\eta_{s,\theta}}{r_0} u_\theta = u_r, \quad (2.15a)$$

$$\Phi_{,t} - B\eta \cos \theta = -\frac{1}{r_0^2} \{2\eta + \cot \theta \eta_{,\theta} + \eta_{,\theta\theta} + \csc^2 \theta \eta_{,\phi\phi}\}. \quad (2.15b)$$

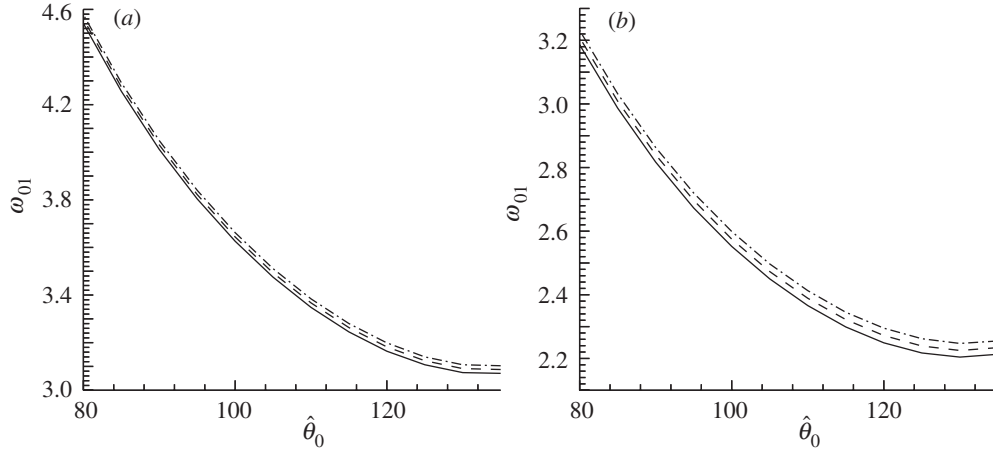


Figure 3. The effect of a small Bond number on the natural frequencies corresponding to the lowest symmetrical modes. $\alpha = 30^\circ$. (a) Liquid in a spherical container; (b) liquid in a cylindrical container. $B = 0$ (solid line), $B = 0.1$ (dashed line) and $B = 0.2$ (dot-dashed line).

Exactly the same procedure that was followed earlier can now be followed to lead to the following infinite set of algebraic equations

$$-\omega \gamma_i a_{mi} = \sum_{j=1}^{\infty} \mu_{ij} b_{mj}, \quad (2.16 a)$$

$$\omega \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = \frac{(\alpha_i + 2)(\alpha_i - 1)}{r_0^2} \gamma_i a_{mi} + \sum_{j=1}^{\infty} \tau_{ij} a_{mj}, \quad (2.16 b)$$

where

$$\tau_{ij} = B \int_0^{\theta_0} \sin \theta \cos \theta P_{\alpha_i}^m P_{\alpha_j}^m d\theta$$

and ν_{ij} and γ_i are as in (2.8); the expression (2.8 b) for μ_{ij} has to be modified to account for the terms involving $B\eta_{s,\theta}$,

$$\begin{aligned} \mu_{ij} = \int_0^{\theta_0} \left[\hat{r}^{\beta_j-1} \{ \beta_j (\cos(\hat{\theta} - \theta) - y \sin(\hat{\theta} - \theta)) P_{\beta_j}^m - (\sin(\hat{\theta} - \theta) + y \cos(\hat{\theta} - \theta)) P_{\beta_j}^{m'} \} \right. \\ \left. - \hat{r}^{-(\beta_j+2)} \left\{ \beta_j (\cos(\hat{\theta} - \theta) - y \sin(\hat{\theta} - \theta)) P_{\beta_j}^m \right. \right. \\ \left. \left. + \frac{\beta_j}{\beta_j + 1} (\sin(\hat{\theta} - \theta) + y \cos(\hat{\theta} - \theta)) P_{\beta_j}^{m'} \right\} \right] \sin \theta P_{\alpha_i}^m d\theta, \end{aligned} \quad (2.16 c)$$

where $y = B\eta_{s,\theta}/r_0$.

Once again we have a matrix eigenvalue problem but now the coefficients a_{mi} cannot be eliminated and z now involves both sets of coefficients.

The effects of a small Bond number on the variation of the lowest frequencies as the volume is reduced is shown in figure 3. In figure 3a, which deals with a spherical-tank supercritical case, the frequency of the symmetrical mode generally increases

with Bond number. The same happens in the 2D cylindrical channel case shown in figure 3*b*. The relevant formulae for this case are given in Appendix A. We have so far only looked for symmetric azimuthal wavenumber modes; the same procedure works for the antisymmetric azimuthal wavenumber modes but we find that the frequencies are identical to the ones found in the corresponding symmetric case.

3. The flat-meniscus case and the other low-Bond-number cases in a spherical tank

Some details of the application of a general procedure to the low-Bond-number supercritical case were presented in § 2. We will now outline the application of the procedure to the other cases shown in figure 1. The only real change needed is the use of different, more appropriate spatial functions in the representation of the interface.

(a) The flat-meniscus, critical-volume case

Let the volume fraction of liquid in the tank be $V_c = 0.75[\frac{2}{3} + \frac{1}{3}\cos^3\alpha - \cos\alpha]$, the critical volume fraction. In this case the static liquid free surface will be flat, as shown in figure 1*e*. Now, it is a surprising and not immediately obvious fact that at this volume fraction *the static interface will be flat for all Bond numbers*. When the interface is flat, the surface tension cannot cause a pressure jump across it and the pressure in the liquid is then purely hydrostatic; and the contact-angle condition is satisfied at all B . This remarkable fact permits us to determine the natural frequencies very accurately for arbitrary B . But first, in order to be able to compare our results with those of McIver (1989), we now, for this subsection alone, use a time-scale $T^* = \sqrt{R/g}$ rather than $T = (\rho R^3/\sigma)$, which was used earlier and which will be used in all the following sections. It can be checked that if ω and ω^* are the frequencies based on T and T^* , respectively, then for $B \neq 0$ or ∞ , $\omega = \sqrt{B}\omega^*$. The two are equal to one another when $B = 1$.

The coordinate system and the functional representation for $\Phi(\hat{r}, \hat{\theta}, \hat{\phi}; t)$ are exactly those used in § 2, and (2.3*b*) is the explicit representation, with (2.4*b*) once again the condition determining the spatial eigenvalues β_j . But now, with a flat interface, the representation for η needs to be different. We choose a cylindrical polar-coordinate system (ρ, ϕ, z) , whose origin is at the centre of the static, flat free surface, z pointing into the gas. We can verify that the contact-angle condition requires that $\partial\eta/\partial\rho$ must vanish on $\rho = \sin\alpha$. Now the two interface conditions in the cylindrical coordinate system are

$$\eta_{,t} = -u_{\hat{r}} \cos \hat{\theta} + u_{\hat{\theta}} \sin \hat{\theta}, \quad (3.1 a)$$

$$\Phi_{,t} + \eta = \frac{1}{B} \left\{ \eta_{,\rho\rho} + \frac{1}{\rho} \eta_{,\rho} + \frac{1}{\rho^2} \eta_{,\phi\phi} \right\}, \quad (3.1 b)$$

where $u_{\hat{r}}$ and $u_{\hat{\theta}}$ are the components of the liquid velocity at the interface in the $\hat{r}$ - and $\hat{\theta}$ -directions, respectively (figure 1*g*). Note that equations (3.1) hold for $0 < B \leq \infty$, i.e. for arbitrary Bond number. Now, whereas Φ continues to be represented in terms of spherical harmonics, it proves convenient to represent $\eta(\rho, \phi, z)$ in terms of a

complete set of cylindrical harmonics

$$\eta(\rho, \phi, z) = \cos \omega^* t \cos m\phi \sum_{j=1}^{\infty} a_{mj} J_m(k_j \rho). \quad (3.2)$$

The infinite set of real scalars $\{k_j, j = 1, 2, \dots\}$ satisfies the condition

$$\left. \frac{d\{J_m(k_j \rho)\}}{d\rho} \right|_{\rho=\sin \alpha} = 0,$$

ensuring that the contact-angle condition is satisfied exactly. Moreover, the representation for η is once again in terms of a set of complete functions. If we now substitute the assumed representations into the interface conditions (3.1), project these onto the basis used for η and eliminate the a_{mi} , we obtain, for $i = 1, 2, \dots$, the infinite system

$$\omega^{*2} \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = \sum_{j=1}^{\infty} \left(1 + \frac{k_i^2}{B}\right) \mu_{ij} b_{mj}, \quad (3.3)$$

where

$$\begin{aligned} \mu_{ij} = \int_0^{\sin \alpha} \rho J_m(k_i \rho) \left[-(\beta_j + m) \hat{r}^{(\beta_j-1)} P_{\beta_j-1}^m \right. \\ \left. + \frac{\hat{r}^{-(\beta_j+2)}}{\beta_j + 1} \{ \beta_j(2\beta_j + 1) \cos \hat{\theta} P_{\beta_j}^m - \beta_j(\beta_j + m) P_{\beta_j-1}^m \} \right] d\rho, \end{aligned} \quad (3.4a)$$

$$\nu_{ij} = \int_0^{\sin \alpha} \rho J_m(k_i \rho) \left[\left(\hat{r}^{\beta_j} + \frac{\beta_j}{\beta_j + 1} \hat{r}^{-(\beta_j+1)} \right) P_{\beta_j}^m \right] d\rho. \quad (3.4b)$$

The system (3.3) is once again of the form $Az = \lambda Bz$ and can as before be solved using standard routines. This case is of some importance for us, since we have the very accurate results of McIver (1989) in tabular form for comparison. McIver's analysis, which uses special coordinate systems, is tailor made for the infinite-Bond-number case, i.e. when surface tension is absent. Table 2 compares the results of our calculations for both the cylindrical tank and spherical tank cases with those of McIver. In both cases the maximum liquid depth is 0.4, corresponding to a contact angle of $\arccos 0.6$. Table 2*a* shows excellent agreement for the 2D symmetric modes with somewhat greater discrepancies for the antisymmetric modes, the maximum being *ca.* 1%. Table 2*b* shows the comparisons for our closed-form results using one or two spatial modes; note with just one mode the error in ω_{s1}^* is less than 2%. The corresponding errors in the antisymmetric frequencies are naturally greater; with one mode the error in ω_{a1}^* is *ca.* 13%, which comes down to 8.6% with just two modes. Table 2*c, d* shows the corresponding comparisons for the spherical tank. Here too we find good agreement, with the greatest discrepancies occurring for the $m = 1$ modes.

We cannot expect the present method, a general method applicable for arbitrary Bond number and different geometries, to be better than, or even to equal, a method

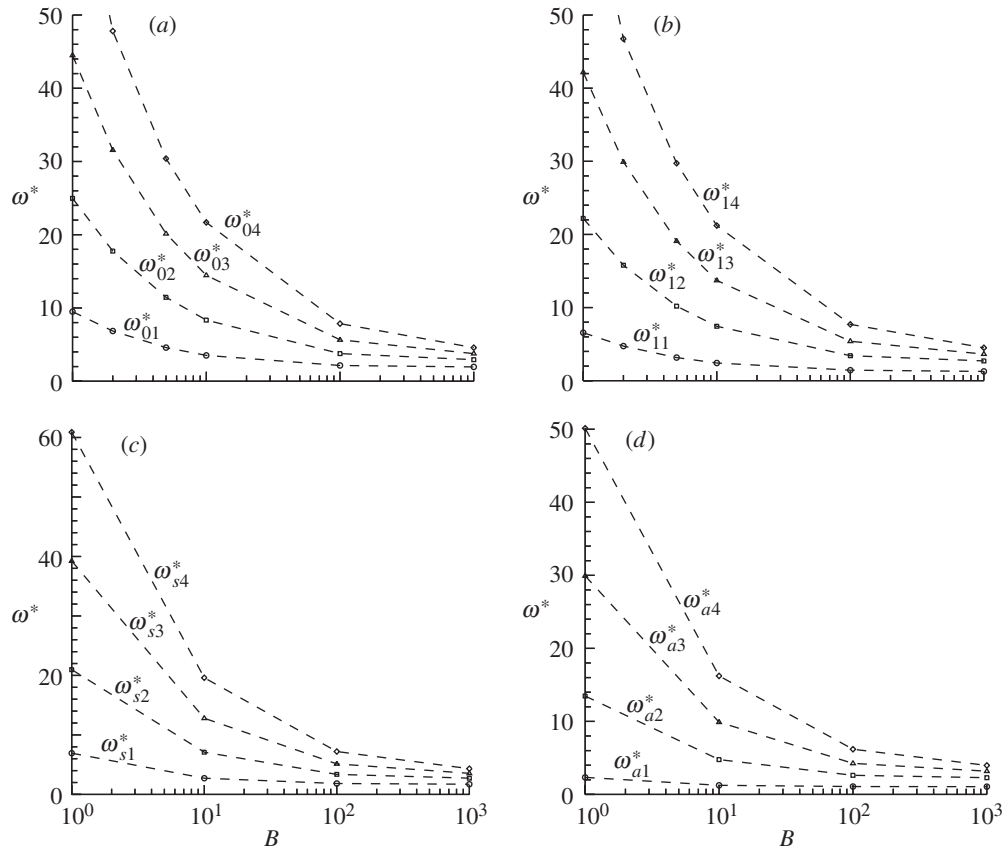


Figure 4. The critical-volume, flat-interface case. The dependence of the lowest frequencies on the Bond number B when the contact angle α is $\arccos 0.6$. (a), (b) A spherical container; (c), (d) a cylinder. Note that here the frequency is non-dimensionalized by $\sqrt{(g/R)}$.

specially designed for the $B = \infty$ flat-meniscus case, for that particular case. However, the power of the present method, already displayed in §2, can once again be seen in the results shown in figure 4 for arbitrary Bond number. All the panels deal with the flat-interface case with $1 \leq B \leq 1000$. As might be expected, since the liquid volume and geometry are fixed, the natural frequencies increase as the Bond number decreases. Note that McIver's method does not handle these cases.

Another check that has been carried out is the computation of the flat-meniscus case for $B = 0$, of course with T as time-scale, and to compare the results with those of the supercritical and subcritical calculations as $V \rightarrow V_c$ from above and below, respectively. As might be expected, the results are consistent.

(b) Subcritical volume of liquid in a spherical tank

We now assume that $V < V_c$. As shown in figure 1*f*, the liquid meniscus will now be convex and the pressure in the liquid will exceed the pressure in the gas. The coordinate system now needed for the interface is shown in figure 1*h* and the

Table 2. Comparison of the results based on the present method with those of McIver (1989) for the flat-interface case, $\alpha = \arccos 0.6 \approx 53.13^\circ$ at infinite Bond number

(Sloshing in a cylindrical container using (a) 50 modes and (b) one or two modes. Sloshing in a spherical container using (c) 15 modes or (d) one or two modes.)

(a)		symmetric modes		antisymmetric modes	
		present	McIver	present	McIver
	ω_1^*	1.700	1.700	1.058	1.047
	ω_2^*	2.644	2.644	2.249	2.222
	ω_3^*	3.317	3.317	3.035	3.001
	ω_4^*	3.869	3.868	3.641	3.603

(b)		symmetric modes			antisymmetric modes		
		$n = 1$	$n = 2$	McIver	$n = 1$	$n = 2$	McIver
	ω_1^*	1.733	1.712	1.700	1.180	1.137	1.047
	ω_2^*	—	2.712	2.644	—	2.392	2.222

(c)		present	McIver		present	McIver
	ω_{01}^*	1.926	1.926	ω_{11}^*	1.097	1.076
	ω_{02}^*	2.817	2.814	ω_{12}^*	2.469	2.382
	ω_{03}^*	3.462	3.456	ω_{13}^*	3.175	3.139
	ω_{04}^*	3.999	3.988	ω_{14}^*	3.844	3.724
	ω_{21}^*	1.495	1.495	ω_{31}^*	1.812	1.812
	ω_{22}^*	2.727	2.724	ω_{32}^*	3.010	3.010
	ω_{23}^*	3.420	3.414	ω_{33}^*	3.656	3.656
	ω_{24}^*	3.974	3.962	ω_{34}^*	4.178	4.179

(d)		$m = 0$			$m = 1$		
		$n = 1$	$n = 2$	McIver	$n = 1$	$n = 2$	McIver
	ω_{01}^*	1.974	1.945	1.926	ω_{11}^*	1.251	1.197
	ω_{02}^*	—	2.888	2.814	ω_{12}^*	—	2.554

interface boundary conditions now read

$$\eta_{,t} + \frac{B\eta_{s,\theta}}{r_0} u_\theta = u_r, \quad (3.5a)$$

$$\Phi_{,t} + B\eta \cos \theta = \frac{1}{r_0^2} \{2\eta + \cot \theta \eta_{,\theta} + \eta_{,\theta\theta} + \csc^2 \theta \eta_{,\phi\phi}\}. \quad (3.5b)$$

Note that once again it is sufficient for the range of $\hat{\theta}$ to be $[0, \hat{\theta}_0]$. As in § 2, both η and Φ are expanded in terms of appropriate spherical harmonics (2.3a) and (2.3b) and projection onto the basis used for η then leads, for $i = 1, 2, \dots$, to the infinite

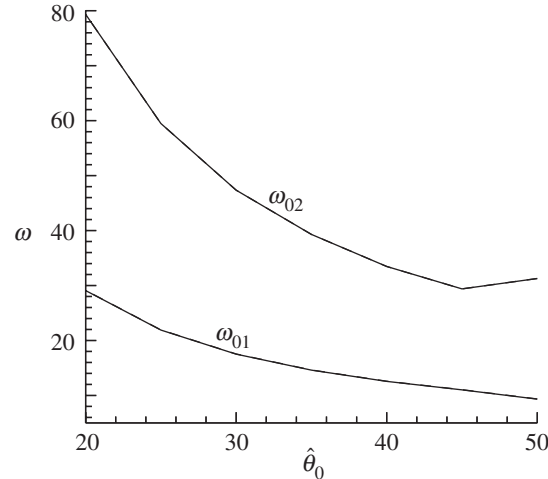


Figure 5. The variation with $\hat{\theta}_0$ of the lowest frequencies corresponding to the symmetrical mode for a subcritical volume of liquid in a spherical container. The contact angle α is $\arccos 0.6$. The solid lines are for $B = 0$. Dashed lines for $B = 0.2$ are also plotted, but these are indistinguishable from the solid lines on this scale.

system

$$-\omega \gamma_i a_{mi} = \sum_{j=1}^{\infty} \mu_{ij} b_{mj}, \quad (3.6a)$$

$$\omega \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = -\frac{(\alpha_i + 2)(\alpha_i - 1)}{r_0^2} \gamma_i a_{mi} - \sum_{j=1}^{\infty} \tau_{ij} a_{mj}, \quad (3.6b)$$

where γ_i, ν_{ij} and τ_{ij} are as in the low- B case of §2, while μ_{ij} has to be modified slightly to account for this geometry.

Figure 5 shows the results of some frequency calculations based on this analysis for the case when $\alpha = \arccos 0.6$. On this graphical scale one cannot see any effect of a small Bond number of 0.2. The frequencies are large now, since the liquid mass becomes small. It can be checked that, as $\hat{\theta}_0 \rightarrow \alpha$, the frequencies tend to the frequencies of the flat-meniscus case considered in the previous subsection.

(c) The centred, off-centre and contacting bubble cases

We finally come to the comparatively simple cases shown in figure 1a–c; since the method is now clear, it is sufficient to consider the zero- g case alone in this subsection. The centred bubble is classical and we mention only that the present method easily works for this case too, with the same coordinate system for both η and Φ . We find, for $i = m, m + 1, \dots$,

$$\omega_{mi} = \sqrt{\frac{(i^2 - 1)(i + 2)}{r_0^3} \left\{ \frac{1 - r_0^{2i+1}}{1 + (i + 1/i)r_0^{2i+1}} \right\}}. \quad (3.7)$$

Strangely, the frequencies are independent of the azimuthal wavenumber. For the off-centre bubble, figure 1b, the same coordinate systems that were used in §2 are

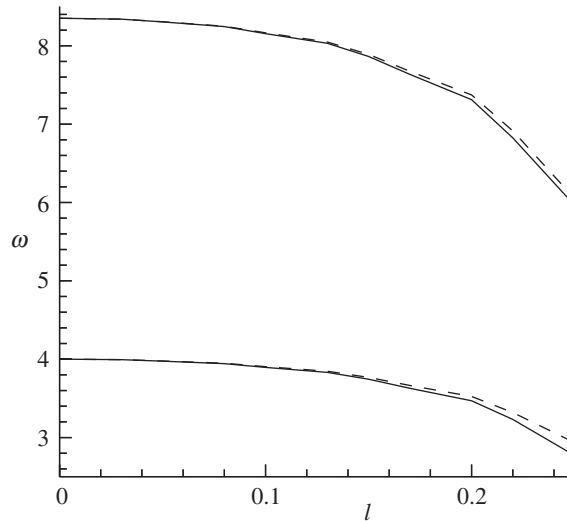


Figure 6. The lowest two natural frequencies for a liquid mass at zero g with an off-centre spherical gas bubble of radius 0.75 in a spherical container. The solid lines are for $m = 0$, while the dashed lines are for $m = 1$. Here l is the displacement of the bubble centre from the centre of the container.

used, and the interface conditions (2.2) once again hold. Although the expansions for $\eta(\theta, \phi; t)$ and $\Phi(\hat{r}, \hat{\theta}, \hat{\phi}; t)$ are once again given by the expressions (2.3), the spherical harmonics needed are now different. There is now no longer a contact-angle condition to be satisfied; rather now smoothness requires that both $\{\alpha_j\}$ and $\{\beta_j\}$ be the set of positive integers. The matrix eigenvalue problem is once again given by (2.7) but there is now a small difficulty that has to be overcome. Since α_1 is now unity, the right-hand side of (2.7b) vanishes for $i = 1$ and there is a simple constraint on b_j that has to be accommodated. Once this is done the analysis goes through as before.

Figure 6 shows how the lowest frequencies change with the position of the gas bubble when the bubble radius is 0.75. The frequencies generally decrease as the bubble becomes more and more off-centre but the change is slow, a useful piece of information from an engineering point of view. The figure also shows another interesting fact. As was pointed out above, with a spherical centred bubble the natural frequencies are independent of the azimuthal wavenumber m . The figure shows that this degeneracy is broken once the bubble is off-centre: although ω_{0i} and ω_{1i} are equal when the displacement $l = 0$, they separate out as l increases. We also observe that in this case, unlike in most other cases, $\omega_{1i} \geq \omega_{0i}$.

When $l = 1 - r_0$ the bubble just touches the tank wall. Now, if $\alpha = 0$, the bubble can stay in position and the analysis just given holds in this case too. If, however, $\alpha \neq 0$, the liquid mass will, on single point contact, redistribute itself so that the contact-angle condition is satisfied. The earlier analyses will then hold. In this manner all the cases shown in figure 1 can be analysed.

4. The extension to arbitrary Bond number

The method advocated in this paper was motivated by low-gravity applications and, in the previous sections, we have shown its application when the Bond number is

small; it is purely by a fortuitous circumstance that it applies, with no extra effort, for all B in the critical-volume case. We will now briefly indicate how the method can be extended to handle the general arbitrary Bond number case for sloshing in a spherical container. We will only consider the supercritical case as an example.

When the Bond number is large, the static liquid interface is no longer spherical or even nearly spherical and this deformation has to be accounted for correctly. The key idea is once again to choose two coordinate systems and they are, for a given α , the same ones that were used for small B , i.e. the spherical polar coordinates indicated in figure 1*g*. Now, however, $r = r_0$ is neither the static interface nor a good approximation to it: but r_0 and O are so chosen that the surface $r = r_0$ passes through the actual static contact line and is tangential to the actual static interface $r = \zeta = \eta_s(\theta)$ there. In this coordinate system the modal frequency equations at the static interface $r = \eta_s(\theta)$ are

$$\eta_{,t} + \frac{\eta_{s,\theta}}{\eta_s} u_\theta = u_r, \quad (4.1 a)$$

$$\Phi_{,t} - B\eta \cos \theta = \kappa_1, \quad (4.1 b)$$

where $\zeta(\theta, \phi; t) = \eta_s(\theta) + \eta(\theta, \phi; t)$ and κ_1 is the extra curvature due to the slosh mode. Unfortunately, the static interface $\eta_s(\theta)$ now has to be computed numerically from the static nonlinear equation governing it and then used, tediously, to compute the interface curvature; we omit these details, which have little to do with the general procedure. The expansions for η and Φ are once again given by (2.3 *a*) and (2.3 *b*), respectively, with the appropriate spherical harmonics again determined by conditions (2.4 *a*) and (2.4 *b*). Note that the contact-angle condition and the no-penetration condition on the tank wall are satisfied exactly. Once again projection onto the basis used for η leads, for $i = 1, 2, \dots$, to the infinite system

$$-\omega \gamma_i a_{mi} = \sum_{j=1}^{\infty} \mu_{ij} b_{mj}, \quad (4.2 a)$$

$$\omega \sum_{j=1}^{\infty} \nu_{ij} b_{mj} = \sum_{j=1}^{\infty} \chi_{ij} a_{mj} + \sum_{j=1}^{\infty} \tau_{ij} a_{mj}, \quad (4.2 b)$$

where χ_{ij} , which involves all the complicated curvature terms resulting from the distorted interface, is given in Appendix C. The calculation of the natural frequencies, however, follows the same procedure as indicated before.

The effects of increasing Bond number on a supercritical liquid volume is shown in figure 7 for a case in which the contact angle is 60° . Here the liquid volume fraction is held constant at 50%. This particular case was chosen for study since it is a case considered in Myshkis *et al.* (1987). Although there is reasonable agreement between the two sets of results, there are systematic discrepancies that cannot be ascribed to scanning and replotting errors alone. Myshkis *et al.* give no details of their calculations and so it is hard to make an accurate assessment. However, we have reason to believe that the discrepancies are due to inaccuracies in the results of Myshkis *et al.* (1987) rather than due to those in ours. First, our calculations agreed very well with the accurate tabulated results of McIver (1989) for the flat-interface case. Second, the present calculations are likely to be very accurate for the $B = 0$ case, where the interface is spherical and the convergence, especially for the $m = 0$

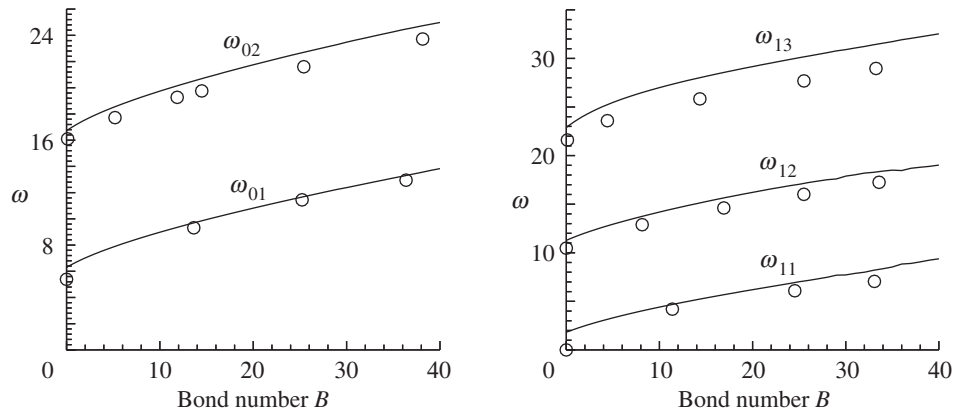


Figure 7. The case of arbitrary Bond number. The variation with Bond number of the lowest natural frequencies of a supercritical volume of liquid in a spherical container. The contact angle is 60° and the liquid volume fraction is held constant at 50%. Solid line, present computations; circles, results of Myshkis *et al.* (1987).

modes, is very rapid. We have used these calculations as a check and find that they converge to the values for $B = 0$ of the arbitrary Bond number procedure rather than to those of Myshkis *et al.* This is of course no proof, but we cannot do much better under the circumstances. Note, however, that the two procedures indicate much the same general trends in a difficult computation.

5. Conclusion

We have suggested a general procedure for the computation of low-gravity slosh frequencies of liquids in tanks of simple shapes. It has been shown that the procedure works for all the cases of interest in spherical and cylindrical tanks. We have also shown that the method can be extended to handle the arbitrary Bond number case, the main nuisance here being the computation of the distorted static meniscus.

It would be in order, in conclusion, to compare the demerits and merits of the present method with other methods available in the literature. A limitation here has been that V could not be too large, large enough for the liquid to occupy the centre of the tank. This is necessary in order to be able to use a simple representation for the potential that permitted easy satisfaction of the normal velocity condition on the tank wall. However, note that in (2.3*b*) the negative powers of $\hat{r}$ become unbounded at the origin and thus representation (2.3*b*) would fail in this case. This restriction can be removed by either using a different representation for Φ or shifting its coordinate system at the expense, however, of simplicity and inaccuracy in satisfying the wall condition. This has in fact been done and the method works well for tanks with large fill volumes. The latter procedure can also be tried if one wants to handle more complicated tank geometries. As far as accuracy is concerned, the method is likely to be as accurate or more so than any other method except those especially tailored for specific geometries and Bond numbers, like that of McIver (1989). The main strengths of this method are, however, its simplicity and generality.

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Appendix A.

Here we write down some of the formulae that pertain to the calculation of low-gravity 2D slosh frequencies in a cylindrical container of circular cross-section.

For the supercritical-volume case, we choose cylindrical polar coordinates (r, θ) and $(\hat{r}, \hat{\theta})$ centred on the mean centre of curvature of the interface and the centre of the cylinder section, for η and Φ , respectively. So figure 1*g* applies to this situation. The interface boundary conditions in the η coordinate system are

$$\eta_{,t} + \frac{B\eta_{s,\theta}}{r_0}u_\theta = u_r, \quad (\text{A } 1 \text{ } a)$$

$$\Phi_{,t} - B\eta \cos \theta = -\frac{1}{r_0^2}\{\eta + \eta_{,\theta\theta}\}. \quad (\text{A } 1 \text{ } b)$$

For slosh modes symmetric in θ , the dependent variables are expanded as follows

$$\eta(\theta; t) = \cos \omega t \sum_{j=1}^{\infty} a_j \cos \alpha_j \theta, \quad (\text{A } 2 \text{ } a)$$

$$\Phi(\hat{r}, \hat{\theta}; t) = \sin \omega t \sum_{j=1}^{\infty} b_j (\hat{r}^{\beta_j} + \hat{r}^{-\beta_j}) \cos \beta_j \hat{\theta}, \quad (\text{A } 2 \text{ } b)$$

where $\alpha_j = j\pi/\theta_0$ and $\beta_j = (2j-1)\pi/(2\hat{\theta}_0)$. The contact-angle condition and the normal velocity condition are satisfied exactly. For the antisymmetric modes, the cosines involving θ in the above have to be changed to sines. Equations (A 1) are now projected onto the basis used for η to obtain, for $i = 1, 2, \dots$, the infinite system

$$-\omega \frac{\theta_0}{2} a_i = \sum_{j=1}^{\infty} \mu_{ij} b_j, \quad (\text{A } 3 \text{ } a)$$

$$\omega \sum_{j=1}^{\infty} \nu_{ij} b_j = \frac{\alpha_i^2 - 1}{r_0^2} \frac{\theta_0}{2} a_i + \sum_{j=1}^{\infty} \tau_{ij} a_j, \quad (\text{A } 3 \text{ } b)$$

where

$$\begin{aligned} \mu_{ij} = \beta_j \int_0^{\theta_0} \cos \alpha_i \theta \left[\hat{r}^{\beta_j-1} \left\{ \cos((\beta_j-1)\hat{\theta} + \theta) + \frac{B\eta_{s,\theta}}{r_0} \sin((\beta_j-1)\hat{\theta} + \theta) \right\} \right. \\ \left. - \hat{r}^{-(\beta_j+1)} \left\{ \cos((\beta_j+1)\hat{\theta} - \theta) - \frac{B\eta_{s,\theta}}{r_0} \sin((\beta_j+1)\hat{\theta} - \theta) \right\} \right] d\theta, \end{aligned} \quad (\text{A } 4 \text{ } a)$$

$$\nu_{ij} = \int_0^{\theta_0} \cos \alpha_i \theta [\hat{r}^{\beta_j} + \hat{r}^{-\beta_j}] \cos \beta_j \hat{\theta} d\theta, \quad (\text{A } 4 \text{ } b)$$

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$$\tau_{ij} = \frac{1}{4}B \sin \theta_0 \left[\cos\{(\alpha_j + \alpha_i)\theta_0\} \left(\frac{1}{\alpha_j + \alpha_i + 1} - \frac{1}{\alpha_j + \alpha_i - 1} \right) + \cos\{(\alpha_j - \alpha_i)\theta_0\} \left(\frac{1}{\alpha_j - \alpha_i + 1} - \frac{1}{\alpha_j - \alpha_i - 1} \right) \right]. \quad (\text{A } 4 \text{ } c)$$

In (A 4), $B\eta_s(\theta)$ is the lowest-order correction to the static meniscus $\zeta = r_0$ caused by the small Bond number and is given by

$$\eta_s(\theta) = \frac{1}{2}r_0^3 [\cos \theta \{1 + \theta_0 \cot \theta_0\} + \theta \sin \theta - \{\cos \theta_0 + \theta_0 \csc \theta_0\}]. \quad (\text{A } 5)$$

When the liquid volume is equal to V_c , the critical volume, there is no change in the coordinate system for Φ but for η we need to use a Cartesian system with origin at the centre of the meniscus. The linearized interface conditions, with T^* as the time-scale, are now

$$\eta_{,t} = u_y, \quad (\text{A } 6 \text{ } a)$$

$$\Phi_{,t} + \eta = \frac{1}{B}\eta_{,xx}. \quad (\text{A } 6 \text{ } b)$$

Now $\eta(x; t)$ is expanded in terms of an appropriate Fourier series in x , while Φ continues to be as in (A 2 *b*). Finally, for the subcritical case the coordinate systems shown in figure 1*h* apply and the analysis proceeds as before.

Appendix B. The curvature in spherical polars and the static interface

In general, the mean curvature of a surface at a point where the unit normal is $\mathbf{n}$ is defined to be

$$\kappa = \pm \nabla \cdot \mathbf{n}. \quad (\text{B } 1)$$

Note that our definition differs by a factor of two from Pozrikidis (1997) and that we will choose the negative sign. In spherical polar coordinates, let the surface be given by $F(r, \theta, \phi) = r - f(\theta, \phi) = 0$. Then the unit normal to the surface is given by

$$\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{r \sin \theta \mathbf{e}_r - \sin \theta f_{,\theta} \mathbf{e}_\theta - f_{,\phi} \mathbf{e}_\phi}{\{(r \sin \theta)^2 + (\sin \theta f_{,\theta})^2 + f_{,\phi}^2\}^{1/2}}. \quad (\text{B } 2)$$

Finally, we have for the mean curvature

$$\begin{aligned} \kappa &= -\nabla \cdot \mathbf{n} \\ &= -\frac{1}{r^2 \sin \theta} \\ &\quad \times \left[\left\{ 2r \sin \theta - \frac{f_{,\phi\phi}}{\sin \theta} - \cos \theta f_{,\theta} - \sin \theta f_{,\theta\theta} \right\} \left\{ 1 + \frac{f_{,\phi}^2}{r^2 \sin^2 \theta} + \frac{f_{,\theta}^2}{r^2} \right\}^{-1/2} \right. \\ &\quad + \left\{ \frac{f_{,\phi}^2}{r \sin \theta} \left(1 + \frac{f_{,\phi\phi}}{r \sin^2 \theta} - \frac{\cot \theta f_{,\theta}}{r} \right) + \frac{\sin \theta f_{,\theta}^2}{r} \left(1 + \frac{f_{,\theta\theta}}{r} \right) + \frac{2f_{,\theta} f_{,\phi} f_{,\theta\phi}}{r^2 \sin \theta} \right\} \\ &\quad \left. \times \left\{ 1 + \frac{f_{,\phi}^2}{r^2 \sin^2 \theta} + \frac{f_{,\theta}^2}{r^2} \right\}^{-3/2} \right]. \quad (\text{B } 3) \end{aligned}$$

Let us say the surface is given by $r = r_0 + \eta(\theta, \phi)$, $\eta \ll r_0$, where r_0 is a constant. Now if (B 3) is linearized we find to leading order

$$\kappa = \frac{2}{r_0} \left[1 - \frac{1}{2r_0} \{ \eta_{,\theta\theta} + \cot \theta \eta_{,\theta} + \csc^2 \theta \eta_{,\phi\phi} + 2\eta \} \right] + O(\eta^2). \quad (\text{B } 4)$$

For a general $r_0(\theta)$ the linearization of (B 3) is quite simple but messy and we omit it here. However, this is what is needed in the arbitrary Bond number case of § 4.

We will now work out the static meniscus when the Bond number is small. The equation for the static meniscus is, from (2.1 b),

$$-B\zeta \cos \theta = \kappa + \gamma. \quad (\text{B } 5)$$

Say $\zeta = r_0 + B\eta_s(\theta)$, where r_0 is a constant; $\kappa = \kappa_0 + \kappa_s$, $\gamma = -\kappa_0$. Then

$$-Br_0 \cos \theta = \kappa_s = -\frac{B}{r_0^2} \{ 2\eta_s + \cot \theta \eta_{s,\theta} + \eta_{s,\theta\theta} \}. \quad (\text{B } 6)$$

Hence, the static meniscus for small B has to satisfy the equation

$$\eta_{s,\theta\theta} + \cot \theta \eta_{s,\theta} + 2\eta_s = r_0^3 \cos \theta. \quad (\text{B } 7)$$

We shall require $d\eta_s/d\theta$ to vanish at $\theta = \theta_0$. If $x = \cos \theta$, the solution is

$$\begin{aligned} \eta_s(\theta) = A'x - \frac{1}{6}r_0^3 \left[x \left\{ \ln(1-x^2) + x^3 \ln \frac{1+x}{1-x} - 2x^2 \right\} \right. \\ \left. + 2(1-x^3) \left\{ \frac{1}{2}x \ln \frac{1+x}{1-x} - 1 \right\} \right], \end{aligned} \quad (\text{B } 8)$$

where the constant A' is given by

$$A' = \frac{1}{3}r_0^3 \frac{\tan(\frac{1}{2}\theta_0)}{\sin \theta_0} [\cos \theta_0 + (1 + \cos \theta_0) \ln \{ 2 \cos^2(\frac{1}{2}\theta_0) \}]. \quad (\text{B } 9)$$

Appendix C. The χ_{ij} term in equation (4.2 b)

It was pointed out in § 4 when dealing with the arbitrary Bond number case that the term χ_{ij} in equation (4.2 b) is complicated by the curvature terms. This term will be explicitly given in this appendix. If the static meniscus is given by $r = \zeta = \eta_s(\theta)$, the perturbed interface will have the representation $r = \zeta(\theta, \phi; t) = \eta_s(\theta) + \eta(\theta, \phi; t)$. Let us define

$$N_1 = -\sin \theta \eta_s (2\eta_s'^2 + 3\eta_s''^2 - \eta_s \eta_s'') + \cos \theta \eta_s' (\eta_s'^2 + \eta_s^2), \quad (\text{C } 1 a)$$

$$D_1 = \sin \theta \eta_s (\eta_s'^2 + \eta_s^2)^{3/2}, \quad (\text{C } 1 b)$$

where the primes denote differentiation with respect to θ . The curvature of the static

interface is then given by $\kappa_0(\theta) = -N_1/D_1$. Now define

$$h_1 = \kappa_0 \frac{\eta_s^2 \sin \theta}{N_1} P_{\alpha_j}^{m''}, \quad (\text{C } 2 \text{ } a)$$

$$h_2 = -\kappa_0 \frac{\eta_s'^2 + \eta_s^2}{N_1 \sin \theta} m^2 P_{\alpha_j}^m, \quad (\text{C } 2 \text{ } b)$$

$$h_3 = \kappa_0 \left\{ \frac{-6\eta_s \eta_s' \sin \theta + (3\eta_s'^2 + \eta_s^2) \cos \theta}{N_1} - \frac{3\eta_s'}{\eta_s'^2 + \eta_s^2} \right\} P_{\alpha_j}^{m'}, \quad (\text{C } 2 \text{ } c)$$

$$h_4 = \kappa_0 \left\{ \frac{(2\eta_s \eta_s'' - 3\eta_s'^2 - 6\eta_s^2) \sin \theta}{N_1} - \frac{\eta_s'^2 + 4\eta_s^2}{\eta_s(\eta_s'^2 + \eta_s^2)} \right\} P_{\alpha_j}^m. \quad (\text{C } 2 \text{ } d)$$

It is understood that in the above the argument of the associated Legendre functions is $\cos \theta$. Then the term χ_{ij} in equation (4.2 *b*) is given by

$$\chi_{ij} = \int_0^{\theta_0} \{h_1 + h_2 + h_3 + h_4\} \sin \theta P_{\alpha_i}^m d\theta. \quad (\text{C } 3)$$

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